

1. Create MariaDB DB on EC2.

- Launched an EC2 instance in the AWS Paris region.
- Installed MariaDB using the DNF package manager.
- Started and enabled the MariaDB service.
- Created a database named **companydb**.
- Created an **employees** table with employee details.
- Inserted sample employee records and verified the data using SQL queries

```
=====
Installed:
mariadb-connector-c-3:3.10-1.amzn2023.0.2.x86_64
mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-errmsg-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-server-3:10.5.29-1.amzn2023.0.1.x86_64
mysql-selinux-1.0.14-2.amzn2023.0.1.noarch
perl-DBI-1.652-1.amzn2023.0.1.x86_64
perl-Math-BigInt-1:1.9998.39-2.amzn2023.0.2.noarch
perl-Math-Complex-1.59-477.amzn2023.0.9.noarch
perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

mariadb-connector-c-config-3:3.10-1.amzn2023.0.2.noarch
mariadb105-backup-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-cracklib-password-check-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-gssapi-server-3:10.5.29-1.amzn2023.0.1.x86_64
mariadb105-server-utils-3:10.5.29-1.amzn2023.0.1.x86_64
perl-DBD-MariaDB-1.22-1.amzn2023.0.5.x86_64
perl-File-Copy-2.34-477.amzn2023.0.9.noarch
perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch
perl-Module-Load-1:0.36-2.amzn2023.0.2.noarch

Complete!
[ec2-user@ip-10-0-0-1-84 ~]$ sudo systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[ec2-user@ip-10-0-0-1-84 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
[ec2-user@ip-10-0-0-1-84 ~]$ sudo mariadb
ERROR 2002 (HY000): Can't connect to local MySQL server through socket '/var/lib/mysql/mysql.sock' (2)
[ec2-user@ip-10-0-0-1-84 ~]$ sudo systemctl start mariadb
[ec2-user@ip-10-0-0-1-84 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-09-17 09:50:41 UTC; 11s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 3983 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 4005 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 4102 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Main PID: 4089 (mariadb)
   Status: "Taking your SQL requests now..."
```

2. Insert some dummy data.

- Created the **employees** table inside the **companydb** database.
- Added sample records containing employee names, departments, and salaries.
- Used the **SELECT** query to verify that the records were inserted successfully.

```

MariaDB [(none)]> SELECT VERSION();
+-----+
| VERSION() |
+-----+
| 10.5.29-MariaDB |
+-----+
1 row in set (0.000 sec)

MariaDB [(none)]> CREATE DATABASE companydb;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> USE companydb;
Database changed
MariaDB [companydb]> CREATE TABLE employees (
  -> id INT PRIMARY KEY AUTO_INCREMENT,
  -> name VARCHAR(100),
  -> department VARCHAR(100),
  -> salary INT
  -> );
Query OK, 0 rows affected (0.008 sec)

MariaDB [companydb]> INSERT INTO employees (name, department, salary)
  -> VALUES
  -> ('Syed', 'DevOps', 50000),
  -> ('Ahmed', 'Cloud', 60000),
  -> ('Rahul', 'Development', 55000);
Query OK, 3 rows affected (0.002 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [companydb]> SELECT * FROM employees;
+-----+-----+-----+-----+
| id | name | department | salary |
+-----+-----+-----+-----+
| 1 | Syed | DevOps | 50000 |
| 2 | Ahmed | Cloud | 60000 |
| 3 | Rahul | Development | 55000 |
+-----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [companydb]>

```

3. Take the backup of dummy data on EC2.

- Created a database backup using the `mysqldump` utility.
- Executed the following command:
- `sudo mysqldump companydb > companydb_backup.sql`
- Successfully generated the `companydb_backup.sql` backup file on the EC2 instance

```
[ec2-user@ip-10-0-1-84 ~]$ sudo mysqldump companydb > companydb_backup.sql
[ec2-user@ip-10-0-1-84 ~]$ ls -lh companydb_backup.sql
-rw-r--r--. 1 ec2-user ec2-user 2.2K Sep 17 09:57 companydb_backup.sql
[ec2-user@ip-10-0-1-84 ~]$ head -20 companydb_backup.sql
/*M!999999\~ enable the sandbox mode */
-- MariaDB dump 10.19 Distrib 10.5.29-MariaDB, for Linux (x86_64)
--
-- Host: localhost      Database: companydb
-- -----
-- Server version      10.5.29-MariaDB

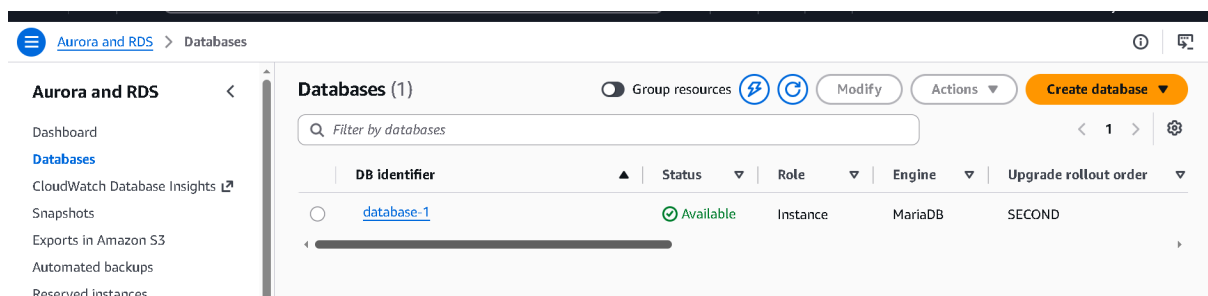
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;

--
-- Table structure for table `employees`
[ec2-user@ip-10-0-1-84 ~]$ sudo mysqldump companydb | head
/*M!999999\~ enable the sandbox mode */
-- MariaDB dump 10.19 Distrib 10.5.29-MariaDB, for Linux (x86_64)
--
-- Host: localhost      Database: companydb
-- -----
-- Server version      10.5.29-MariaDB

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
mysqldump: Got errno 32 on write
[ec2-user@ip-10-0-1-84 ~]$
```

4. Launch MariaDB RDS instance.

- Created an Amazon RDS MariaDB database in the AWS Paris region.
- Configured the RDS instance inside the project VPC.
- Used private connectivity and configured the required security group.
- Enabled secure SSL-based database connections



5. Migrate database from EC2 to RDS.

- Downloaded the Amazon RDS SSL certificate bundle.
- Connected to the MariaDB RDS instance using the MariaDB client.

- Created the **companydb** database on RDS.
- Imported the EC2 database backup into RDS using the following command:

```
mariadb -h <RDS-ENDPOINT> -u admin -p companydb < companydb_backup.sql
```

- Verified that the database and employee records were successfully migrated

```
[ec2-user@ip-10-0-1-84 ~]$ mysql -h database-2.cnaeao2kodo1.eu-west-3.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 83
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| innodb |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.006 sec)

MariaDB [(none)]> |
```

```
MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| innodb |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.006 sec)

MariaDB [(none)]> CREATE DATABASE companydb;
Query OK, 1 row affected (0.008 sec)

MariaDB [(none)]> EXIT;
Bye
[ec2-user@ip-10-0-1-84 ~]$ mariadb -h database-2.cnaeao2kodo1.eu-west-3.rds.amazonaws.com -P 3306 -u admin -p --ssl-ca=./global-bundle.pem --ssl-verify-server-cert companydb < companydb_backup.sql
Enter password:
[ec2-user@ip-10-0-1-84 ~]$ mariadb -h database-2.cnaeao2kodo1.eu-west-3.rds.amazonaws.com -P 3306 -u admin -p --ssl-ca=./global-bundle.pem --ssl-verify-server-cert companydb
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 89
Server version: 11.8.8-MariaDB-log managed by https://aws.amazon.com/rds/

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [companydb]> SHOW TABLES;
+-----+
| Tables_in_companydb |
+-----+
| employees |
+-----+
1 row in set (0.001 sec)

MariaDB [companydb]> SELECT * FROM employees;
```

6. Install MySQL DB on EC2.

- Installed Docker on the EC2 instance.
- Downloaded the MySQL 8.0 Docker image.
- Created and started a MySQL container.
- Mapped EC2 port **3307** to MySQL container port **3306**.
- Connected to MySQL using the Docker command.
- Created the **testdb** database and inserted sample employee records

```

2026-09-17T11:35:16.292833Z 0 [System] [MY-013602] [Server] Channel mysql_main conf
2026-09-17T11:35:16.298497Z 0 [Warning] [MY-011810] [Server] Insecure configuration
insider choosing a different directory.
2026-09-17T11:35:16.339666Z 0 [System] [MY-011323] [Server] X Plugin ready for conn
2026-09-17T11:35:16.339933Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready
MySQL Community Server - GPL.
[ec2-user@ip-10-0-1-84 ~]$ sudo docker exec -it mysql-server mysql -u root -p
Enter password:
ERROR 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)
[ec2-user@ip-10-0-1-84 ~]$ sudo docker stop mysql-server
mysql-server
[ec2-user@ip-10-0-1-84 ~]$ sudo docker rm mysql-server
mysql-server
[ec2-user@ip-10-0-1-84 ~]$ sudo docker run -d \
  --name mysql-server \
  -e MYSQL_ROOT_PASSWORD='MySQL@12345' \
  -p 3307:3306 \
  mysql:8.0
875e6b0666f3804c0c93b89a9805fcba12cb626c86bfeb693cda2f17ca80fd80
[ec2-user@ip-10-0-1-84 ~]$ sudo docker exec -it mysql-server mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql>

```

7. Launch MySQL RDS image.

- Created a MySQL RDS instance named `mysql-rds-project`.
- Selected MySQL Community Edition with the `db.t4g.micro` instance class.
- Configured the database in the project VPC in the Paris region.
- Selected private access and configured the required security group.
- Waited until the RDS status changed to **Available**.

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> CREATE DATABASE testdb;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> USE testdb;
Database changed
```

```
mysql>
mysql> CREATE TABLE employees (
  ->     id INT AUTO_INCREMENT PRIMARY KEY,
  ->     name VARCHAR(50),
  ->     department VARCHAR(50),
  ->     salary INT
  -> );
INSERT INTO employees (name, department, salary)
Query OK, 0 rows affected (0.03 sec)
```

```
mysql>
mysql> INSERT INTO employees (name, department, salary)
  -> VALUES
  -> ('Syed', 'DevOps', 50000),
  -> ('Ahmed', 'Cloud', 60000),
  -> ('Rahul', 'Development', 55000);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> SELECT * FROM employees;
+-----+-----+-----+-----+
| id | name  | department | salary |
+-----+-----+-----+-----+
| 1  | Syed  | DevOps    | 50000  |
| 2  | Ahmed | Cloud     | 60000  |
| 3  | Rahul | Development | 55000  |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql>
```

```
A newer release of "Amazon Linux" is available.
Version 2023.12.20260914:
Run "/usr/bin/dnf check-release-update" for full release and version update info

#
#####
# Amazon Linux 2023
#
#####
# https://aws.amazon.com/linux/amazon-linux-2023
#
#####

Last login: Thu Sep 17 11:58:11 2026 from 103.203.172.230
[ec2-user@ip-10-0-1-84 ~]$ curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 165408 100 165408 0 0 2667k 0 --:--:-- --:--:-- --:--:-- 2692k
[ec2-user@ip-10-0-1-84 ~]$ mariadb -h mysql-rds-project.cnaea2kodo1.eu-west-3.rds.amazonaws.com -P 3306 -u admin -p --ssl-ca=./global-bundle.pem --ssl-verify-server-cert
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 8.4.9 Source distribution

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.043 sec)

MySQL [(none)]> exit;
Bye
[ec2-user@ip-10-0-1-84 ~]$
```

8. Configure Multi-AZ.

RDS tasks have been completed. However, Task 8 is partially completed because my AWS Free Tier account is not eligible for the required Multi-AZ configuration.

9. Take backup of DB and restore the DB.

- Created a backup of the MySQL `testdb` database using `mysqldump`.
- Saved the backup as `testdb_backup.sql`.
- Created a new database named `restored_db`.
- Restored the backup into the new database using the MySQL client.
- Verified that the employee records were successfully restored.

```
ec2-user@ip-10-0-1-84:~
[ec2-user@ip-10-0-1-84 ~]$ sudo docker exec mysql-server mysqldump -u root -p'MySQL@12345' testdb > testdb_backup.sql
mysqldump: [warning] Using a password on the command line interface can be insecure.
[ec2-user@ip-10-0-1-84 ~]$ ls -lh testdb_backup.sql
-rw-r--r--. 1 ec2-user ec2-user 2.1K Sep 17 11:45 testdb_backup.sql
[ec2-user@ip-10-0-1-84 ~]$
```

10. Create read replica.

- Connected to the MySQL RDS instance from the EC2 server.
- Downloaded the RDS SSL certificate bundle.
- Used an SSL-secured connection to access the RDS database.
- Verified the connection by running the `SHOW DATABASES;` SQL command.
- Successfully established connectivity between EC2 and MySQL RDS

Aurora and RDS > Databases > Create read replica

ⓘ

🔍

Account

598122633072

KMS key ID

26b4de79-3efb-441d-a04e-c44b375a3196

Maintenance

Auto minor version upgrade [Info](#)

☒ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version [documentation](#).

☐ Enable deletion protection

Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

Cancel

Create read replica